Math153LogisticRegressionModel

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```r
library(UsingR)
```

```
## Loading required package: MASS

## Loading required package: HistData

## Loading required package: Hmisc

##
## Attaching package: 'Hmisc'

## The following objects are masked from 'package:base':
##
##     format.pval, units
```

```r
library(ggplot2)
library(Bolstad)
```

```
##
## Attaching package: 'Bolstad'

## The following objects are masked from 'package:stats':
##
##     IQR, sd, var
```

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:Hmisc':
##
##     src, summarize

## The following object is masked from 'package:MASS':
##
##     select

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(Bolstad2)
```

```
##
```

```
## Attaching package: 'Bolstad2'

## The following object is masked from 'package:Bolstad':
##
##     sintegral

## The following object is masked from 'package:Hmisc':
##
##     describe
```

```r
library(rstanarm)
```

```
## Loading required package: Rcpp

## This is rstanarm version 2.32.2

## - See https://mc-stan.org/rstanarm/articles/priors for changes to default priors!

## - Default priors may change, so it's safest to specify priors, even if equivalent to the defaults.

## - For execution on a local, multicore CPU with excess RAM we recommend calling

##   options(mc.cores = parallel::detectCores())
```

```r
library(bayesplot)
```

```
## This is bayesplot version 1.15.0

## - Online documentation and vignettes at mc-stan.org/bayesplot

## - bayesplot theme set to bayesplot::theme_default()

##    * Does _not_ affect other ggplot2 plots

##    * See ?bayesplot_theme_set for details on theme setting
```

```r
library(broom)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v stringr   1.6.0
## v lubridate 1.9.4     v tibble    3.3.0
## v purrr     1.2.0     v tidyr     1.3.1
## v readr     2.1.5

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter()    masks stats::filter()
## x dplyr::lag()       masks stats::lag()
## x dplyr::select()    masks MASS::select()
## x dplyr::src()       masks Hmisc::src()
## x dplyr::summarize() masks Hmisc::summarize()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
theme_set(theme_classic())
library(tidyverse)
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
```

```
##     lift
##
## The following objects are masked from 'package:rstanarm':
##
##     compare_models, R2
```

```
library(GGally)
library(ggplot2)
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
library(bayesplot)
theme_set(bayesplot::theme_default(base_family = "sans"))
library(rstanarm)
options(mc.cores = 1)
library(loo)
```

```
## This is loo version 2.9.0
## - Online documentation and vignettes at mc-stan.org/loo
## - As of v2.0.0 loo defaults to 1 core but we recommend using as many as possible. Use the 'cores' ar
```

```
library(projpred)
```

```
## This is projpred version 2.10.0.
```

```
SEED=14124869
```

```
#loading full dataset
babies_df <- babies
```

Cleaning Dataset

```
babies_df_somewhat_reduced = subset(babies_df, wt != 999 & ht != 99 & parity != 99 & age != 99 & smoke!
```

Creating the binary variable

```
babies_df_w_bin <- babies_df_somewhat_reduced %>%
  mutate(uw = case_when(
    wt < 88.1849 ~ 1,
    wt > 88.1849 ~ 0
  ))
```

gestation, ht, wt1, number, parity, age parameters

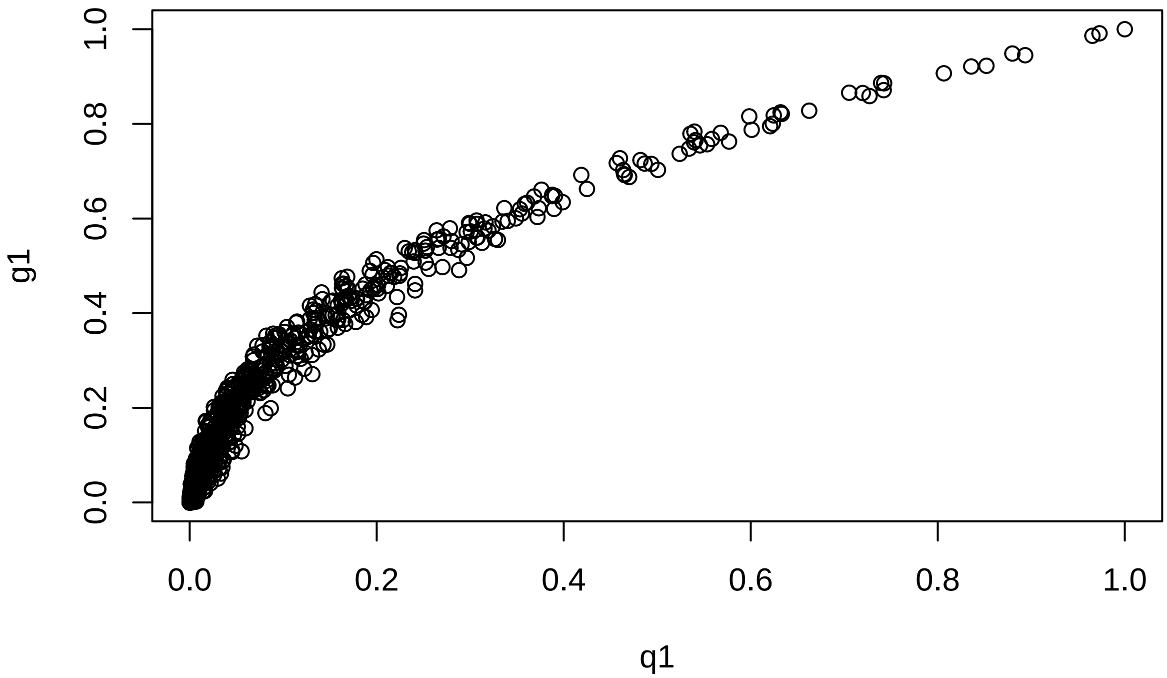
```
#with the gestation, ht, wt1, number, parity, age parameters
#making a matrix of the gestation, ht, wt1, number, parity, age parameters

babies_reduced4 <- babies_df_w_bin %>%
  dplyr::select(uw, wt, gestation, ht, wt1, number, parity, age)
```

```
babies_reduced4_matrix <- data.matrix(babies_reduced4)
```

One Metropolis Hastings sample

```r
log <- BayesLogistic(babies_reduced4_matrix[, 1], babies_reduced4_matrix[, c(3,4,5,6,7,8)])
```

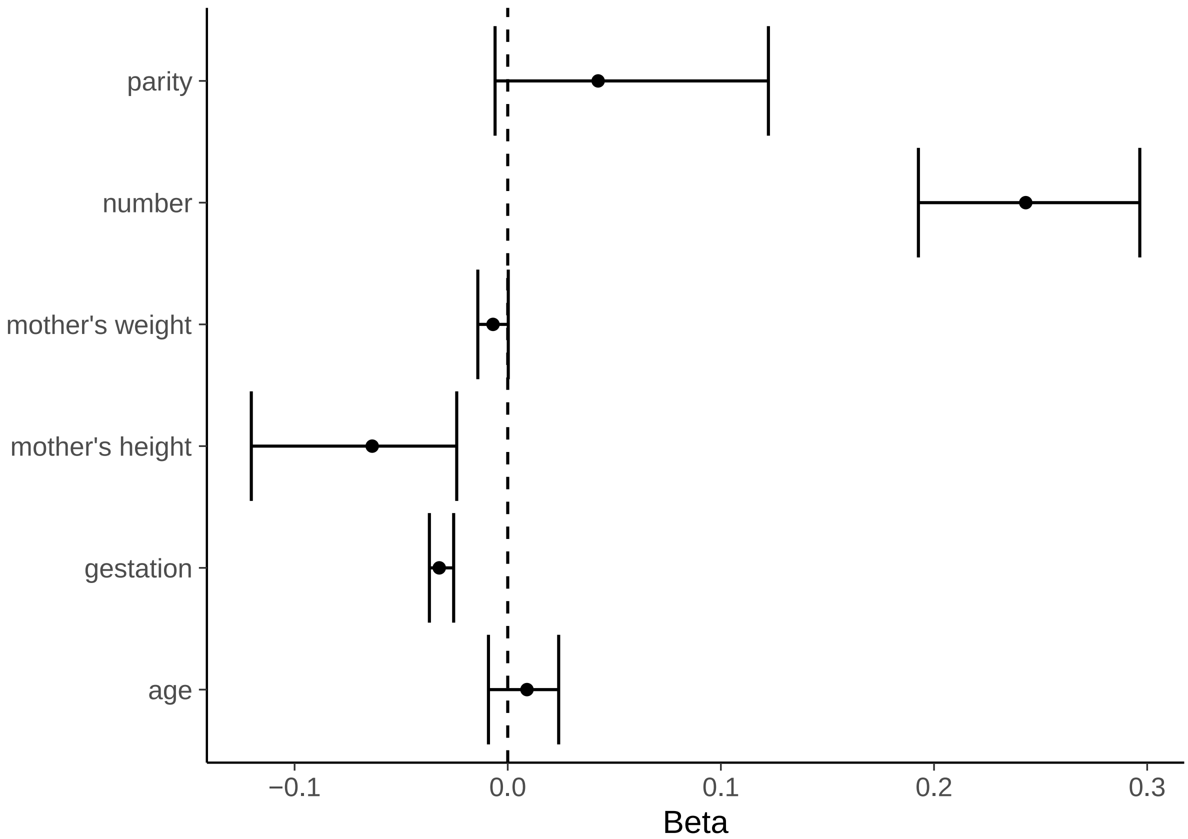



```
##       N         mean       stdev         sterr          min           q1
## b0 1000  9.386921474 3.713542055 0.1174325108 -2.70273945  6.617689545
## b1 1000 -0.031710070 0.007071302 0.0002236142 -0.05349358 -0.036550950
## b2 1000 -0.062821820 0.057222957 0.0018095488 -0.21622220 -0.105093917
## b3 1000 -0.003621957 0.007232500 0.0002287117 -0.02867064 -0.008486398
## b4 1000  0.224595802 0.057485988 0.0018178665  0.01414942  0.184378917
## b5 1000  0.024436944 0.080953880 0.0025599865 -0.21613015 -0.025390491
## b6 1000  0.012870505 0.026054850 0.0008239267 -0.05126777 -0.006382541
##            med           q3          max
## b0  9.43364105 11.871274911 21.629809383
## b1 -0.03153511 -0.027069987 -0.008167235
## b2 -0.06140139 -0.022612627  0.149542078
## b3 -0.00372264  0.001417584  0.018844088
## b4  0.22510033  0.263433239  0.409276707
## b5  0.02801628  0.078676931  0.262792902
## b6  0.01270657  0.032028618  0.080431800
##       Mean.beta StdDev.beta     Z.beta
## b0  9.386921474 3.713542055  2.5277542
## b1 -0.031710070 0.007071302 -4.4843329
## b2 -0.062821820 0.057222957 -1.0978429
## b3 -0.003621957 0.007232500 -0.5007891
## b4  0.224595802 0.057485988  3.9069660
## b5  0.024436944 0.080953880  0.3018625
## b6  0.012870505 0.026054850  0.4939773
```

```r
beta <- c("gestation", "mother's height", "mother's weight", "number", "parity", "age")
beta_means <- c(-0.032133756, -0.063594586, -0.006912550, 0.243027690, 0.042408552, 0.009020142)
beta_lower_ci <- c(-0.036757896, -0.120308926, -0.014037164, 0.192669370, -0.005946375, -0.009043999)
beta_upper_ci <- c(-0.0253790046, -0.0239433079, 0.0002454796, 0.2965387412, 0.1222946333, 0.0238970350)

beta <- data.frame(beta, beta_means, beta_lower_ci, beta_upper_ci)
```

```r
beta %>%
  ggplot(aes(beta_means, beta)) +
  geom_point() +
  geom_errorbar(aes(xmin = beta_lower_ci, xmax = beta_upper_ci))+
  geom_vline(aes(xintercept=0), lty=2)+
  xlab("Beta") +
  ylab("")
```



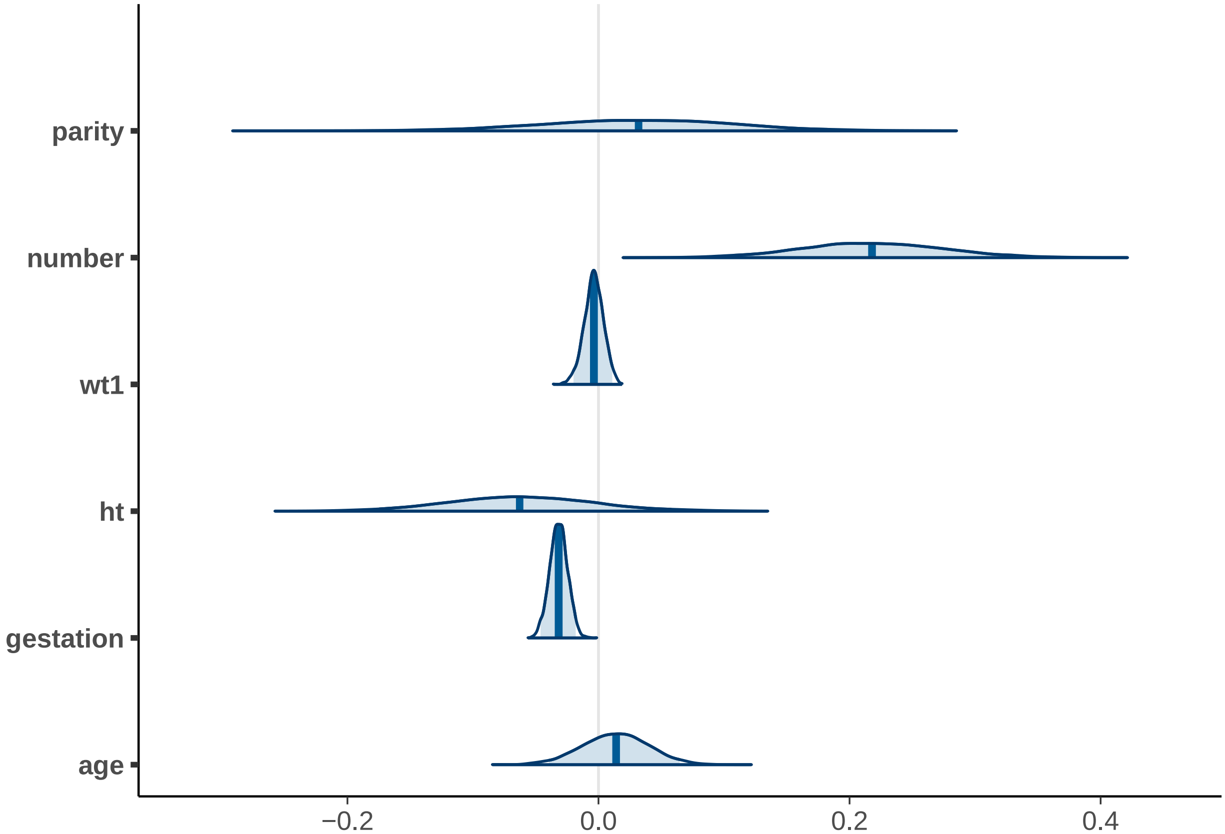
Using a new function for simplicity!

```r
#using weak prior
t_prior <- student_t(df = 7, location = 0, scale = 2.5)

#
post1 <- stan_glm(uw ~ gestation + ht + wt1 + number + parity + age, data = babies_reduced4,
                  family = binomial(link = "logit"),
                  prior = t_prior, prior_intercept = t_prior, QR=TRUE,
                  seed = SEED, refresh=0)
```

```r
post1_mat <- as.matrix(post1)
```

```r
mcmc_areas(post1_mat,
           pars = c("parity", "number", "wt1", "ht", "gestation", "age"),
           prob = 0.95)
```




```r
round(coef(post1), 2)
```

```
## (Intercept)   gestation          ht         wt1      number      parity 
##        9.38       -0.03       -0.06        0.00        0.22        0.03 
##         age 
##        0.01
```

```r
round(posterior_interval(post1, prob = 0.95), 5)
```

```
##                 2.5%    97.5%
## (Intercept)  1.88380 16.69715
## gestation   -0.04615 -0.01815
## ht          -0.17436  0.05759
## wt1         -0.01991  0.01100
## number       0.10748  0.33160
## parity      -0.12405  0.18223
## age         -0.03935  0.06468
```